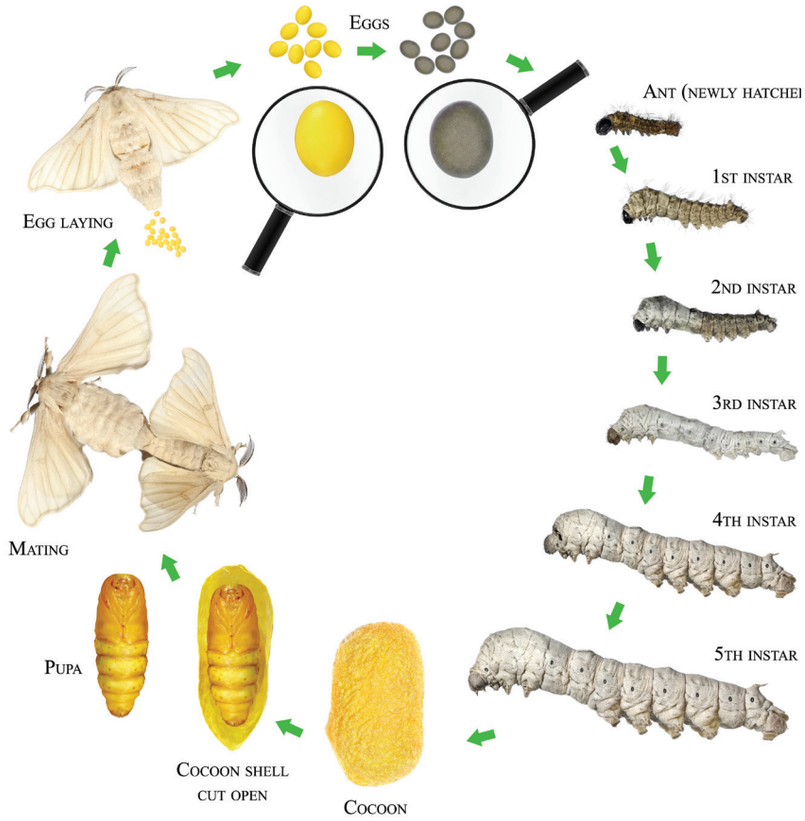


Figure 13.3 Life cycle of *Bombyx mori*

about 3 mm in length and is pale, yellowish-white in colour. The caterpillars are provided with well developed mandibulate type of mouth-parts adapted to feed easily on the mulberry leaves.

After 1st, 2nd, 3rd and 4th moultings caterpillars get transformed into 2nd, 3rd, 4th and 5th instars respectively (Figure13.3). It takes about 21 to 25 days after hatching. The fully grown caterpillar is 7.5 cm in length. It develops salivary glands, stops feeding and undergoes pupation. The caterpillars stop feeding and move towards the corner among the leaves and secretes a sticky fluid through their silk gland. The secreted fluid comes out through spinneret (a narrow pore situated on the hypopharynx) and takes the form of long fine thread of silk which hardens on exposure to air and is wrapped around the body of caterpillar in the form of a covering called as cocoon. It is the white coloured bed of the pupa whose outer threads are irregular while the inner threads are regular. The length of continuous thread secreted by a caterpillar for the formation of cocoon is about 1000-1200 metres which requires 3 days to complete. The pupal period lasts for 10 to 12 days and the pupae cut through the cocoon and emerge into adult moth.

On the basis of the moults which they undergo during their larval life, *B. mori* is divided into three races – tri-moulters, tetra-moulters and penta-moulters. Based on voltinism (the number of broods raised per year), three kinds of races are recognized in mulberry silkworm – univoltines (one brood only), bivoltines (two broods only) and multivoltines (more than two broods).

India has the distinction of producing all the four types of silk i.e. (a) Mulberry silk (91.7%); (b) Tasar silk (1.4%); (c) Eri silk (6.4%); and (d) Muga silk (0.5%) which are produced by different species of silkworms. Name the species that produces large amount and least amount of silk in India.

Cultivation of food plants for the silkworms

The first component, is to grow the food plants for the silkworms. Mulberry leaves

XI Std Zoology Chapter-13 EM.indd 242 30/12/2022 14:08:29